

CODE Beyond FAIR: A Roadmap for Reusable Research Software

Simple Hinglish Summary (Scientific Data Journal, 2026) — for explaining to Sir

1. Ye Paper Kis Baare Mein Hai? (What is this paper about?)

Sir, **ye paper research software (yaani scientists/researchers dwara likha gaya code) ko better banane aur reuse karne ke liye ek roadmap deta hai.** Aajkal science mein sirf paper aur data hi nahi, software bhi ek important research output ban gaya hai — kai research articles mein software ka use hota hai (33% se badh kar 48% ho gaya hai 2013 se 2021 tak).

2. Problem Kya Hai? (What is the Problem?)

- FAIR principles (**Findable, Accessible, Interoperable, Reusable**) **pehle sirf data ke liye bane the, software ke liye nahi.**
- Software, data se alag hota hai — ye **living nature** ka hota hai, matlab ye continuously change/update hota rehta hai (jaise stable version, beta version, deprecated version, sab saath rehte hain).
- Isliye software ke liye ek final **"perfect version" nahi hota** jise archive kar dein — **balki poori history record karni padti hai.**
- Bahut sa research software share hi nahi hota**, ya sahi tarike se **document nahi hota**, jisse dusre researchers usko reuse nahi kar paate.

3. Authors Ka Solution: CODE Roadmap

Authors ne ek simple aur step-by-step roadmap banaya hai jiska naam hai **CODE** — iske 4 parts hain: **Open, Document, Execute, Collaborate.** Har part mein kuch cheezein "mandatory" (zaroori), kuch "recommended" (suggested) aur kuch "optional" hain — taaki har scientist, chahe wo coding expert ho ya na ho, dheere-dheere apna software improve kar sake.

(A) OPEN — Code ko Public Karna

- Publish karo (mandatory):** Apna code ek public website (jaise GitHub, GitLab) par daalo, sirf email par manga kar mat do — kyunki email/website change ho sakta hai.
- Archive karo (mandatory):** Code ko ek permanent jagah save karo, jaise "Software Heritage" naam ki ek non-profit organization, jo hamesha ke liye software preserve karti hai.
- License do (strongly recommended):** Bina license ke, koi bhi legally aapka code use nahi kar sakta, chahe aap chahte ho ki log use karein. Isliye open-source license dena zaroori hai.
- Authorship batao:** Kaunse authors/team ne code banaya, ye clear hona chahiye.

(B) DOCUMENT — Code Samjhaana

- Variables aur functions ke **meaningful names** rakho (jaise "da" ki jagah "discrete_analysis").
- Code mein **comments** likho jo batayein "kyun" ye code aisa likha gaya, sirf "kya" nahi.
- Examples/tutorials** do taaki naye users aasani se seekh sakein.
- Bade software ke liye poori **API documentation** honi chahiye.

(C) EXECUTE — Software Chalaana Aasan Banao

- Apne software ke saare **dependencies** (jo aur software/tools chahiye) list karo, taaki koi aur bhi usko run kar sake.
- Ek **computational environment** do (jaise Docker container) taaki result reproduce ho sake.
- **Test suite** banao — chhote examples jinka sahi answer pata ho, taaki verify ho sake ki software sahi kaam kar raha hai.
- Real-life **usage examples** do expected results ke saath.

(D) COLLABORATE — Milkar Kaam Karna

- Users ke **issues/feedback** ka reply do.
- Batao ki contributions accept karte ho ya nahi, aur kaise contribute karein.
- Apni **maintenance limits** clearly bataao — kitna support de sakte ho.
- Ek **community** banao — workshops, tutorials, hackathons karke.

4. Sirf Researchers Ki Zimmedari Nahi

Paper kehta hai ki sirf researchers/scientists par bharosa nahi karna chahiye — **institutions, funders, libraries, aur publishers** ko bhi apna role nibhana chahiye:

- **Institutions:** Software developers ko hire karein, software banane ko career mein importance dein, apne khud ke forges (platforms) banayein.
- **Funders:** Sirf naya software banane ke liye nahi, balki purane accha software maintain karne ke liye bhi grant dein (kyunki bahut zaroori software sirf ek-do logon ke free time par depend karta hai — jaise NumPy library).
- **Libraries:** Software ko documents/articles ki tarah archive aur catalog karein.
- **Publishers:** Research paper ke saath code publish karna mandatory karein, sirf "available upon request" kaafi nahi hai.

5. Conclusion (Nichod)

Research software ab science ka ek bahut important hissa ban chuka hai, lekin abhi tak ise papers/data jitni importance nahi milti. Ye CODE roadmap ek simple, step-by-step tarika deta hai jisse har scientist — chahe wo coding expert ho ya beginner — apna software share, document, execute aur collaborate ke liye better bana sake. Sath hi, institutions, funders, libraries aur publishers ko bhi is process ko support karna chahiye, taaki science zyada reproducible aur reliable ban sake.

Quick Cheat-Sheet (Sir ko dikhane ke liye)

Category	Kya Karna Hai (Mandatory items only)
OPEN	Code public forge par publish karo + dedicated archive mein save karo
DOCUMENT	Meaningful names + comments (recommended)
EXECUTE	Dependencies list karo (recommended)
COLLABORATE	Issues ka reply do (recommended)

Reference: Di Cosmo, R., Granger, S., Hinsén, K., Jullien, N., Le Berre, D., Louvet, V., Maumet, C., Maurice, C., Monat, R. & Rougier, N. P. (2026). CODE beyond FAIR: a roadmap for reusable research software. Scientific

